

EDUCATION

MSc Computer Science

ETH Zürich

09/2020 – 11/2024

Zurich, CH

Courses

- Info Security Lab (5.75), German C1 (5.75), System Security (5.75), Digital Signatures (5.75), Program Analysis for Security (5.75), Computational Statistics (5.25)

CMS & BSc Computer Science

EPFL

09/2016 – 09/2020

Lausanne, CH

Cégep

Dawson College

08/2014 – 06/2016

Montréal, CA

SKILLS

Python (PyTorch)

SQL

React

Java

REST

Blender

LANGUAGES

English

Native / Bilingual

German

Full Professional

French

Full Professional

Arabic

Professional

AWARDS

Peter Berlow Award

Dawson College

- Highest average in first year of Applied Sciences.

AI for Good Hackathon

Entrepreneur First, Visium

- 1st place, organized by Entrepreneur First, Visium & EPFL AI Team.

Best Project – EcoHackathon

ETH AI Center, Gainforest

- Bird species ID from acoustic data via ML (wav2vec2 on Xeno Canto).

Amro Abdrabo

MSc Computer Science, ETH Zürich

MSc in Computer Science from ETH Zürich and BSc in Computer Science from EPFL, with solid and versatile CS experience.

EXPERIENCE

Graduate Assistant (PhD)

HEC - UNIL

02/2025 – Present

Assistant and Researcher

- Teaching assistant for the Python programming and Algorithms course, instructing 500+ students. Responsible for maintaining the data-driven surveys website: datadrivensurvey.com. Published in ACM IMWUT. Courses: doplab.unil.ch/act/ people.unil.ch/kevinhuguenin/teaching/info2/

Master's Thesis

Irmos Technologies AG

Zurich, Switzerland

Structural Monitoring

- Applied unsupervised AI methods for continuous monitoring of bridge structural conditions using vibration time-series data. Built a physical bridge model in ANSYS WB to determine modal shapes. Published in Sensors MDPI.

Fullstack Developer

ETH Juniors

10/2022 – 01/2023

Zurich, Switzerland

Photogrammetric application development

- Full stack in C#, JS, HTML, Python, Docker, Azure. Reduced backend processing time by ~23%. Designed architecture and identified security risks for guest registration. Developed geometric calculation methods and used statistical techniques to improve 3D models.

Assistant

EPFL

09/2019 – 12/2019

Lausanne, Switzerland

Teaching Assistant

- Teaching assistant in Analysis at EPFL (Prof. Dr. Anna Lachowska).

Summer Internship in Brain Imaging

Dawson College & Brain Imaging Centre

05/2015 – 08/2015

Montréal, Canada

MRI Signal Processing

- Phase correction for MRI signal acquisition under Prof. Dr. Nadeau.

SELECTED PROJECTS

Ethereum Smart Contract Security Analysis

02/2022 – 06/2022

- Security analysis of smart contracts using 1,000 lines of Datalog. Grade: 6.0. Source: github.com/AmroAbdrabo/ethereum-sc-taint-analyzer

VOJ – Voice of Jungle

ETH AI Center Hackathon Winner – 05/2024

- Identifies bird species from acoustic data using ML. Trained wav2vec2 transformer on Xeno Canto data. Demo: youtu.be/tnENCnPxDO

OpenEye

Entrepreneur First Hackathon Winner – 04/2024

- Early disease detection via AI-powered retinal scanners with medical data anonymization. Winner of Visium's 'AI for Good'.

Catrine

02/2022 – 06/2022

- 3D puzzle game built with MonoGame (C#) and C++ physics engine. Trailer: youtu.be/WluS6iGwTIA

Electricity Consumption Forecasting

Datathon – 05/2024

- Third-lowest error forecasting ALPIQ electricity consumption in Spain and Italy.

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